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Department of Design, Analytics and Cyber Security
T. Y. BSc Cyber and Digital Science
Academic Year 2025–2026
Information Security policy and Audit

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**CUSTOMER STORY:BARCLAYS' QUANTUM-READY
TRANSFORMATION WITH CRYPTOMATHIC**

Abstract

Barclays, a global financial services leader, partnered with Cryptomathic to address escalating cryptographic costs, complex siloed **hardware security module (HSM)** infrastructure, and the emergent threat of quantum computing. Cryptomathic's **Crypto Service Gateway (CSG)** provided a centralized cryptographic management solution, enabling Barclays to achieve significant cost savings, enhance cryptographic agility, and establish a future-proof, quantum-ready security infrastructure. This transformation freed up budget for post-quantum initiatives and drastically reduced deployment timelines

Key words

Cryptomathic, Barclays, Crypto Service Gateway (CSG), Quantum-ready, Post-quantum cryptography (PQC), Cryptographic agility, Hardware Security Modules (HSMs), Cost savings, Digital transformation.

Introduction

Barclays, a global financial services leader with over 300 years of legacy, has consistently prioritised innovation in banking technology. This includes being an early adopter of digital transformation, with a strong emphasis on cryptographic security to ensure scalable, cost-efficient operations and prepare for future threats, particularly from quantum computing. This document details Barclays' strategic move to enhance its cryptographic posture through a partnership with

Background

Barclays has a history of innovation in banking, from launching the UK's first credit card to installing the world's first ATM. Operating in over 50 countries, the bank has a proactive stance towards digital transformation, focusing on both current efficiency and future readiness.

Cryptographic security is a foundational element of its strategy.

Scenario

Barclays faced several significant challenges in managing its cryptographic operations:

- **Escalating Costs:** Managing hundreds of siloed hardware security modules (HSMs) resulted in rising infrastructure costs and inefficiencies.
- **Complexity and Inefficiency:** Fragmented systems led to delays in cryptographic updates and operations, hindering project delivery and innovation.
- **Quantum Threats:** The impending threat from quantum computing necessitated future-proofing the infrastructure to secure sensitive operations.
- **Limited Agility:** Adapting to new cryptographic standards or integrating post-quantum algorithms was cumbersome, creating readiness gaps. Cryptographic agility, the ability to update algorithms and switch vendors without disruption, was identified as essential for resilience against immediate threats and future innovations.

Objectives

Barclays' objectives in addressing these challenges, leading to their partnership with Cryptomathic, included:

- Centralising cryptographic management to reduce infrastructure complexity and inefficiencies across its global operations.
- Reducing escalating cryptographic costs and optimising HSM utilisation.
- Enhancing cryptographic agility to allow quick adaptation to evolving security needs, update algorithms, and integrate new HSMs or standards seamlessly.
- Preparing for quantum computing threats by building a future-proof, quantum-ready security infrastructure capable of integrating post-quantum algorithms.

IS Audit Standards The provided sources do not contain specific information about IS audit standards that were applied or referenced during this transformation project.

Key Findings

The partnership between Barclays and Cryptomathic, through the implementation of the Crypto Service Gateway (CSG), yielded transformative results:

- **Reduced Costs:** Barclays significantly cut hardware costs by centralizing operations and optimising HSM usage, freeing up budget for investment in post-quantum initiatives.
- **Enhanced Agility:** Cryptographic bottlenecks were eliminated, reducing deployment timelines from months to weeks. This allowed Barclays to adapt quickly to changing requirements, update algorithms, and integrate new HSMs.
- **Future-Ready Infrastructure:** Barclays is now equipped with a scalable, modular platform that supports post-quantum cryptography and ensures security for decades to come. CSG was specifically designed for seamless integration with post-quantum algorithms, ensuring a smooth transition to quantum-safe operations.
- **Global Scalability:** The CSG framework operates seamlessly across Barclays' global network, managing cryptographic operations from central hubs in the UK, Singapore, and the United States.
- **Operational Simplification:** Unified cryptographic policies simplified compliance processes and reduced audit costs.

Conclusion

Barclays' adoption of Cryptomathic's Crypto Service Gateway demonstrates a successful strategy for achieving both immediate and long-term benefits through cryptographic agility. By reducing costs, centralising operations, and proactively preparing for quantum computing threats, Barclays has solidified its position as a leader in secure and innovative financial services. This investment not only simplifies the path to post-quantum cryptography but also creates value today by enabling faster, more efficient cryptographic management. The solution addressed Barclays' challenges of escalating costs, complexity, quantum threats, and limited agility, resulting in a future-proof security infrastructure and significant operational improvements.

References

- [Collection of Case Studies - Cryptomathic](#)

07/08/2025

Date of Submission

Sign of Instructor